Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam D · Form 1: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. An incline block tied to a hanger over an ideal pulley slides uphill but slows down. The rope is taut. What are the directions of kinetic friction and acceleration on the block?

(A) Both uphill

(B) Friction up, acceleration down

(C) Both downhill

(D) Friction down, acceleration up

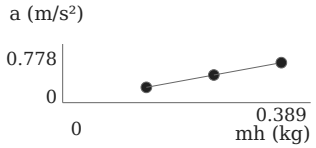
2. Transfer mass between a frictionless cart and its hanger on an ideal pulley. Total mass is 5 kg; g=10 m/s². Double the fixed total mass while using the same hanger masses. Under the revised model, what is Y at X=0.2688?

(A) -0.5712

(B) 1.529

(C) 0.2688

(D) 0.6888



3. A descending hanger is pulling a cart when the hanger reaches a shelf and stops. The rope then goes slack. Which previous relation must no longer be imposed?

(A) Equal acceleration magnitudes for cart and hanger

(B) Constant gravitational field

(C) Newton's second law on the cart

(D) Downward weight on the hanger

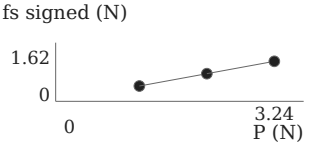
4. An incline block of weight 10 N rests on a 30° ramp with a 5 N hanger on an ideal pulley. Add vertical downward force P to the block. Friction is sufficient; uphill is positive. Change the tilt so sinθ increases by a factor of 1.5. Static friction remains sufficient. What is the new ratio of vertical change to the same horizontal change?

(A) 0.54

(B) 1.065

(C) 0.75

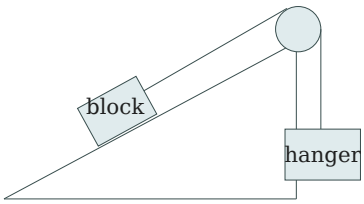
(D) 0.855



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Downhill block weight component is 12 N and hanger weight is 10 N. The block slides uphill against 3 N kinetic friction. Total system mass is 4 kg.

a. Draw kinetic friction on the incline block. **(1 pt)**



b. Calculate acceleration now and the static friction needed for rest immediately after stopping. **(3 pt)**

6. Release a 60 kg cabin and adjustable counterweight over an ideal pulley without a motor. Cabin up is positive; g=10 m/s². acabin (m/s²) -> Tcable (N): (1.24, 674.4); (2.24, 734.4); (3.24, 794.4).

a. Build the model for counterweight elevator from physics and predict both slope and intercept. **(2 pt)**

b. Halve cabin mass and produce the same cabin acceleration values. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A cart of mass M slides toward a pulley against kinetic friction μMg. A hanger of mass m descends. Derive a condition for the hanger to slow down while descending. **(3 pt)**